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The foundations of 3D-MOT as a cognitive enhancement tool

Les fondements du 3D-MOT en tant qu'outil d'amélioration cognitive

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Abstract

The domain of cognitive enhancement is vast, spanning behavioral, biochemical, and physical applications. The techniques are as numerous as are the limitations: poorly conducted studies, ethically ambiguous practices, limited positive effects, significant side-effects, high financial costs, significant time investment, unequal accessibility, and lack of transfer.

This article provides empirical foundations and supporting evidence for the use of a novel tool – 3-dimensional multiple object tracking – in a cognitive enhancement context. An overview of current literature for cognitive enhancement, as well as the foundations upon which this tool is based, are presented herein. Proposed domains for enhancement include information processing, attention, working memory, inhibition, and executive functions.

Improvements have been demonstrated in attention, working memory and visual information processing in healthy adults; for biological motion processing in healthy aging subjects; on-field performance in soccer players; and in attention for a population with a neurodevelopmental deficit. More work in these and in other fields remains a requisite for further use and development of this technique.

Keywords: cognitive enhancement • information processing • attention • working memory • inhibition • executive functions • ecological validity • 3-dimensional multiple object tracking

Résumé

Le domaine de l'amélioration cognitive couvre des applications comportementales, biochimiques et physiques.

Les techniques sont aussi nombreuses que les limites : études mal menées, pratiques éthiquement ambiguës, effets bénéfiques limités, effets secondaires importants, coûts financiers élevés, investissement en temps important, accessibilité inégale et souvent sans preuve de transfert.

Cet article fournit des fondements empiriques et des preuves à l'appui de l'utilisation d'un nouvel outil – le suivi tridimensionnel d'objets multiples – dans le contexte d'amélioration cognitive. Un survol de la littérature scientifique sur l'amélioration cognitive, ainsi que les fondements sur lesquels repose cet outil, est présenté. Les domaines proposés pour une amélioration comprennent le traitement d'information, l'attention, la mémoire de travail, l'inhibition et les fonctions exécutives.

Des améliorations ont été démontrées dans l'attention, la mémoire de travail et le traitement de l'information visuelle chez des adultes en bonne santé ; pour le traitement du mouvement biologique chez des sujets vieillissants en bonne santé ; de la performance sur le terrain chez les joueurs de soccer ; et en attention pour une population présentant un déficit neurodéveloppemental. Plus de travail dans ces domaines et dans d'autres demeure une condition préalable à l'utilisation et au développement ultérieur de cette technique.

Mots clés : amélioration cognitive • traitement d'information • attention • mémoire de travail • inhibition • fonctions exécutives • validité écologique • 3-dimentional multiple object tracking

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Introduction

Psychological interventions have traditionally been restricted to clinical practices targeting abnormal functions, or high-performance domains (for example business, sports) in the hopes of cultivating these high-level abilities. In recent years, there has been a marked increase in interventions that apply to healthy and normally functioning individuals; an emerging field called *cognitive enhancement*. “Cognitive enhancement is defined as the improvement or strengthening of mental skills including attention, memory, processing speed, and problem holding.” [1] Long is the list of purported *cognitive enhancers*, “interventions in humans that aim to improve mental functioning beyond what is necessary to sustain or restore good health.” [2]

Many activities in daily life have been demonstrated to improve cognitive capacities, to a more-or-less important degree. Examples include reading, knitting, gardening, playing a musical instrument [3], and playing board/card games [4, 5]. Other research has demonstrated neuroplastic effects for activities such as complex navigation [6], juggling [7] and memory training [8]. Domains spanning pharmacology, nutrition, exercise, sleep, meditation, mnemonics, brain stimulation, computer training, video games, and neurofeedback, to name a few, have all been said to enhance cognitive function.

Dresler and colleagues [9] separate cognitive enhancement strategies into three large, overlapping domains: biochemical (for example, pharmaceuticals), physical (for example, neurostimulation) and behavioural (for example, meditation). Many of the existing techniques are bimodal, for example a neural implant could be considered both physical and biochemical. This paper presents the framework for one of the latter, a behavioural intervention called 3-dimensional multiple object tracking, or 3D-MOT, as a cognitive enhancer.

Cognitive enhancement

The field of cognitive enhancement via behavioural modalities arguably began with puzzle-like pencil-paper games, and these types of methods are still a mainstay in traditional academic practices. Newer technologies allow for virtual tools, computer-based training programs and video games being the two major categories, differentiated most often by the medium and user interface (for example console and gamepad, computer mouse and keyboard) as well as in terms of the degree of adaptability and personalization [1].

Rather than divide by presented medium and user interface, most scientific research makes the distinction between two types of video games; those designed primarily for entertainment purposes, *video games*, and those created with the primary purpose of enhancing cognitive functions, *serious games*. In the case of the former, entertainment value is typically formulated as “enjoyment and sustained

player engagement” [10]. In regard to serious games, the focus is “challenging underlying neural systems or specific cognitive abilities” [10]. These games often lack the immersive elements central to the entertainment value of video games [10]. Video games and computer training are considered attractive options for cognitive enhancement as they are readily available, often require little specialized equipment, and are often self-motivating. [11].

Video games

The first generation of video games were not unlike computerized cognitive training, in that they are repetitive, of increasing difficulty and are often comprised of tasks not unlike some neuropsychological tests. Tetris is a wonderful example of this, and in 1994 research demonstrated improvement in mental rotation and spatial visualization following gamelan [12]. Entertaining video games could potentially, at least according to popular media and marketing campaigns, improve attention, visuospatial skills, information processing speed, and executive control [1].

Serious games

Early video games like Tetris loosely resemble existing products designed with the specific purpose of enhancing cognitive functions, called *serious games*, or *computerized brain-training programs*. These programs are defined as interventions targeting a core cognitive skill, including but not limited to attention, memory, processing speed, executive functions, motor functions, emotional regulation. The intervention often consists of discrete, targeted, repetitive activities of increasing difficulty which seek improvement in a cognitive domain by way of enhancing neural functioning [1].

As a large field, computer-based cognitive training has shown that it is possible to enhance memory [13, 14, 15], working memory [16], attention [14], executive function [17], and processing speed [18].

However, the evidence that these programs result in wider-ranging benefits, possible because of *transfer*, is often lacking. Studies that examine these effects often report little evidence of transfer beyond the context of training [19]. At the same time, the sheer amount of commercialized computer brain-training programs makes any generalization extremely difficult, other than to say that more research is needed [1, 10, 11]. Still, this hasn't seemingly disrupted a market that has significantly grown into a major industry with some titans producing far-reaching consumer products [20].

3-Dimensional Multiple Object Tracking (3D-MOT)

3D-MOT can be classified as a behavioural intervention and as a serious video game. We'll first briefly discuss the applications of 3D-MOT as a cognitive enhancer.

Experience playing a video game increases MOT performance, suggesting a functional link between video games and MOT [21]. Research demonstrates that those who

regularly play video games are able to track an average of two more items in the MOT task than non-gamers [21]. Those who train on certain types of video games (involving abilities similar to MOT, for example 'first person shooters') demonstrate a variety of improved abilities in the MOT task [22], and expert gamers demonstrate faster information processing speed abilities in the MOT paradigm than non-gamers [23].

Interestingly, these studies used MOT to demonstrate transfer following video-game training but did not address the possibility that MOT itself could be a cognitive enhancer. One research team ultimately noted that the MOT task resembled one of the training paradigms which led to cognitive enhancement [22], and another study noted that with repeated MOT testing, task performance improved [23].

Subsequent research has suggested that training with 3D-MOT has been demonstrated to enhance attention, working memory and visual information processing speed in healthy participants [24], attention in those with neurodevelopmental deficits [25], on-field performance in athletes [26], and memory [27] and biological motion processing in healthy aging participants [28].

Critics argue that the benefits of 3D-MOT training are weak (small effect sizes) in domains of working memory and sustained attention, often lack evidence of ecologically valid transfer, and are completely devoid of preregistered studies. Some studies were also weakened by a small *n*, variable methodology, and inadequate control groups [29]. Further, a recent study of 3D-MOT in a healthy young adult population fail to show any transfer, near (defined as a second object tracking task) or far (route monitoring; cited as an ecologically valid measure of multitasking ability), although the study does report some improvements in working memory [30].

■ The origin of Multiple Object Tracking (MOT)

The original Multiple Object Tracking (MOT) tool was developed by Plyshyn and Storm [31] as a research tool to investigate a theory of visual indexing, the FINST Indexing Theory. The task has been adapted in many ways and operates as both an evaluation and training tool, based on context.

The FINST Indexing theory hypothesizes that there is a cognitive mechanism responsible for 'indexing' visual objects (or features of objects) for subsequent cognitive processes. The MOT tool was thus designed to examine the locus/loci of attention and address the following questions: does attention work as a single beam which is shifted from location to location? Or can attention be divided across multiple targets [31]?

An MOT paradigm is thus comprised of multiple 'targets' and identical 'distractors'. Stimuli are most often presented on or projected onto a screen. The identities of each are revealed to the subject in one of the first steps of the task by breaking the homogeneity of the objects. Following a trigger, the subject must, in their minds, 'paint' or 'tag'

the identities of targets and distractors onto the identical objects, and then follow them through a delimited space over a given period of time, finally identifying the targets at the end of the trial [31].

The MOT research tool has revealed many properties of attention in a tracking context. Targets can be tracked if occlusion occurs [32], when all objects briefly disappear from view (for example, during an eye-blink) [33, 34], and into the visual periphery [35]. Tracking is viewed as an on-line process, with new information being refreshed and old information suppressed; initially distinguishing features are 'deleted' and replaced by more current information [36]. The degree of speed, occlusion and of change in direction of objects correlates negatively with tracking accuracy [37, 38]. Speed might not be a limiting factor in MOT performance, however [39]. Interestingly, the attentional 'paint' process and the inhibitory 'suppress' process appear at least partially independent of one another, the latter being less limited than the former [40]. Object tracking across visual hemifields can be processed independently in each cortical hemisphere [41]. Tracking four to five targets appears to be the upper limit of MOT [31], however there are contexts in which it is possible to track an even larger number [37].

The idea that MOT can be used as a training tool is born of the observation that task performance increases over time [22], both for repeated [42] and novel trajectories [43]. By adapting the task to a 3-dimensional virtual environment, ecological validity crises [44].

As discussed above, improvements in 3-dimensional MOT (3D-MOT) benefit elderly populations, somewhat reversing the effects of age-related deficits in biological motion perception [28]. In an athletic context, complex scene dynamics demand a high level of performance of these systems, and athletes also appear to benefit from 3D-MOT training in this context [44]. 3D-MOT abilities can assess post-operative cognitive dysfunctions in open-heart surgery patients [45], can predict post-concussion status years after a minor traumatic brain injury [46], and has the potential to rehabilitate affected cognitive functions [45, 47].

Variables in the MOT paradigm that are subject to manipulation include the number, shape, colour and size of objects (targets and distractors) [37], their speed [39], their distribution across the visual field (location) [41], their direction and pattern of movement [37], the number of spatial dimensions (2D [31] or 3D [44]), and finally, the duration of movement [48].

■ Method: The 3D-MOT paradigm

A specific type of MOT paradigm has since emerged. A complete overview of the 3D-MOT paradigm methodology is presented below. *Figure 2* presents a visual representation of the 5 different steps in a traditional 3D-MOT task trial.

- a) Presentation: eight spheres appear randomly dispersed in a virtual cube, homogeneous and coloured yellow. At the center of the virtual cube is a neutral fixation point, a red dot. This phase lasts two seconds.
- b) Indexation: the four target spheres turn red, with a surrounding white halo. This lasts two seconds. The four target spheres then return to their original colour to restore homogeneity for an additional second.
- c) Movement: This phase lasts eight seconds. All eight targets move along a linear path in the virtual 3-D cube. Initial movement direction is randomized. If a sphere comes in contact with another sphere or a wall of the cube, it bounces off and resumes its trajectory. If a sphere passes behind another, occlusion occurs. During tracking the subject's eyes must remain focused on the neutral fixation point while tracking occurs primarily in the peripheral visual field.
- d) Stoppage and identification: All eight spheres cease movement and are labeled with numbers (1 to 8). Subjects verbally state their responses. The answers are input by a trainer using a keyboard, and the selected targets are identified with a halo. The verbal response ensures that subjects are not encumbered and so they do not need to move their focus from off the screen. The subjects also have a video-game controller that allows them to rotate the cube and make the foreground spheres transparent in order to reveal any spheres that might be hidden behind another sphere in the 3-D virtual space. This phase ends when the subject validates their response by pressing a button on the controller.
- e) Feedback: The target spheres are revealed, and feedback (target spheres correctly and incorrectly identified) is given. This phase lasts two seconds.

If a subject correctly identifies all four target spheres, the speed of movement increases for the subsequent trial. If all four target spheres are not identified, the speed of movement decreases for the subsequent trial. The changes in speed are variable and follow an adaptive staircase. The design of the adaptive staircase is such that larger variations (up to 100%) occur in initial trials (to more quickly reach the appropriate

speed) and smaller variations (as low as 10%) in later trials (to maintain the zone of proximal development). For each trial, the speed and number of correct targets identified can be recorded, and it is even possible to record trial object variables such as number of interactions (impacts, occlusions). At the end of 20 trials, a threshold speed score is displayed for subjects and recorded in a database.

The duration and intensity of a given training session can vary, but a session is generally broken down into three identical series comprised of 20 trials each. A session's threshold score is the average of the thresholds of the three series. The amount of time spent training in each session is approximately 30 minutes. Sessions are traditionally performed between 9 am and 5 pm on non-consecutive days at a frequency of two or three times per week. The total number of sessions performed can vary, but in much of the peer-reviewed research performed to date, subjects perform a minimum of 10 training sessions [24-26, 28]. In research and clinical studies, it is suggested to begin a 3D-MOT training session with a questionnaire evaluating level of fatigue, recent sleep, and other potentially relevant variables (for example, consumption of psychiatric medication and psychoactive substances) to control for aberrant performance results.

Materials/Apparatus

In a research context, 3D-MOT sessions are often performed in the C.A.V.E. (Cave Automatic Virtual Environment) [49]. The C.A.V.E. is a 10 foot by 10 foot by 10 foot enclosure onto which is projected the 3D-MOT task. The MOT environment consists of a large cube measuring approximately 1.5 meters in length, width, and height, while targets measure 10 cm diameter. The participant is seated at a distance of 1.5 meters from the screen and is given a fixation point located in the center of the cube. With this setup, the 3D-MOT task utilises a visual field of approximately 45 degrees. The use of a cube as the environment allows for horizontal and vertical movement of both targets and distractors to remain roughly equivalent. Speed is measured in meters per second (m/s);

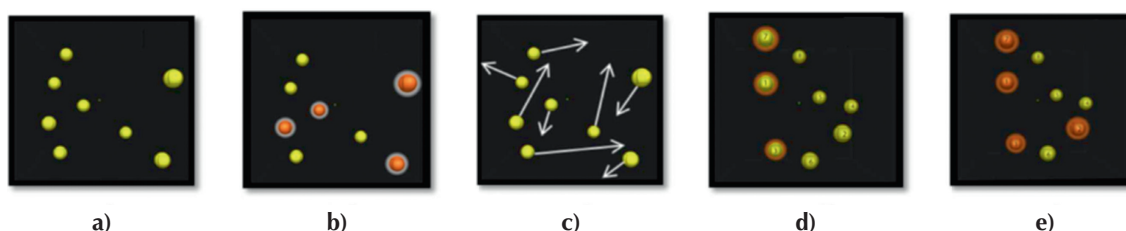


Figure 1. Visual representation of a 3D-MOT trial
Adapted from Faubert & Sidebottom [44].

each trial begins with either a fixed initial start speed (for example, 0.3), or an adaptive speed based on previous performance (for example, 10% below previous speed threshold). The 3-D aspect of the MOT task is achieved using stereoscopic projection and active shutter lenses synchronized to 120Hz. Commercially available 3D-MOT products, such as NeuroTracker [50], allow the task to be performed on a regular computer monitor or tablet, using anaglyph 3D glasses, although the quality of 3D and the size of the visual field are highly variable because of non-standardized equipment and methods.

Discussion

The brief history of MOT as a cognitive enhancer

The notion that MOT could be used as a cognitive enhancer is a fairly recent one, potentially born of the observations that with repeated MOT resting, task performance improves [23] and because of the similarities between MOT and other cognitive enhancement paradigms [22].

As with other cognitive enhancers, in healthy, normal functioning adults (for example, college and university students), 3D-MOT has been shown to improve performance in attention, working memory and visual information processing speed [24]. In healthy aging participants, transfer to the domain of biological motion perception has also been demonstrated [28]. In athletes (soccer players) who are already considered superior performers at the 3D-MOT task [51], impressively, a complex outcome and multivariate transfer measure of 'on field performance' improves [26]. In a clinical population with neurodevelopmental deficits, attention is improved [25].

Cognitive functions involved in 3D-MOT

While 3D-MOT is a complex task requiring several cognitive functions, there is a significant body of research which seeks to tease out the roles of each of these abilities within the task. It is important to understand these functions, as these are the potential domains in which 3D-MOT could potentially act as a cognitive enhancer.

Mentioned above, the initial goal of MOT tasks was to better understand attentional loci [31], and it can thus be said to be principally an attentional task in the visual modality. A larger view of the task quickly reveals other functions involved, including visual information processing [31, 35, 38], attention [31], inhibition [40], working memory [24, 32], executive functions [24, 26, 52]. Each of these functions and their role in the 3D-MOT task will be discussed.

Visual information processing

The ability to perceive, discriminate and identify visual stimuli is a good working definition of visual information processing [53]. The processing of visual stimuli is fundamental

for the identification and tracking of objects and is considered the lowest-level cognitive function implicated by the task [54].

The visual system in the human brain is a complex neural network that, once beyond the confines of the primary visual cortex, can be simplified and divided into two major pathways. Up until fairly recently, these were categorized as the ventral 'object vision' involved in recognition and identification, or *what*, stream; and a dorsal 'spatial vision' involved in localization and tracking, or *where*, stream [55]. More recently, some researchers have begun to consider another role of the dorsal stream; *how*. This system would thus also be responsible for transforming information in an "egocentric frame of reference in preparation for direct action", thus participating in the coordination of behaviours involving specific spatial locations [56].

While these pathways are but two of the many networks involved in processing visual information, it is the latter, dorsal, pathway that is likely of the most interest with regard to the MOT task [57]. Posterior parietal areas, notably the intraparietal sulcus and the superior parietal lobule, make up the principal components of the '*where/how*' visual system [58].

Norman [59] provides an interesting description of the differences between these two systems, a summary of which is presented in *table 1* below. Most important are the characteristics of the dorsal stream, which lend themselves extremely well to the MOT task. The contribution of the dorsal stream to visually guided behaviour, the fact that it is 'motion oriented' and utilises 'a low-level' of conscious visual processing, at high speed, well into the periphery, make it an obvious candidate for the 3D-MOT paradigm, and research tends to confirm this [60].

In addition to being involved in information processing, the same neuroanatomical regions of the medial parietal cortex are also involved in visual attention and visual memory [61], the former of which will be discussed next, and the latter in a later section.

Attention

A strangely intuitive but intangible state, attention has been the subject of intense focus (the irony notwithstanding) and debate since William James' seminal text on the subject [62]. Attention is the subject of countless definitions, subtypes and theoretical models; auditory, visual, visuospatial, sustained, selective, divided, distributed, dynamic, multifocal, conscious, unconscious, executive, active, passive, bottom-up, and top-down are all ways of describing attention and its varieties. It is easy to get lost in the bottomless pit of attentional concepts.

One thing is certain: attention is one of the functions at the very core of cognitive functioning [62, 76] and is also central to the 3D-MOT task [31]. Visual attention is required for identification of targets and on-line updating of their spatial location, and attention must be distributed dynamically throughout the course of the MOT task [54].

Table 1. Characteristics of the ventral and dorsal visual pathways.

	Ventral ‘what’ system	Dorsal ‘where/how’ system
Function	Recognition/identification	Visually guided behaviour
Sensitivity	High spatial frequencies; detail-oriented	High temporal frequencies; motion oriented
Memory	Long term	Very short term
Speed	Relatively slow	Relatively fast
Consciousness	Typically high-level	Typically low-level
Frame of reference	Object-centered (allocentric)	Viewer-centered (egocentric)
Visual input	Mainly foveal and parafoveal	Across retina (fovea and periphery)
Monocular vision	Generally small effects	Often large effects (motion parallax)

Adapted from Norman [59]

A particular focus on selective, sustained, and distributed attention, primarily in the visual modality, will be made. In a separate section, the role of inhibition in attention, and more specifically with the 3D-MOT task will be discussed.

Selective attention is colloquially referred to as *focus*; it is the capacity to attend to a given stimulus or stimuli [63]. It can be thought of analogously as a beam emitted from a flashlight directed to a specific stimulus. In the case of 3D-MOT, selective attention is the focus applied to target spheres. Selective attention is partially a ‘bottom-up’ driven process, meaning that it is mediated by unconscious factors such as salience and object features, as well as partially ‘top-down’, with conscious, willful intent driven elements [56].

Neuroanatomically, the bottom-up processes involve the visual stream up to the primary visual cortex, to V1, which contain small receptive fields, simple feature identification; and to V4, where large receptive fields and complex feature identification are found. Beyond the visual cortex, areas involved in selective attention include the posterior parietal cortex, within which is found the intraparietal sulcus, which itself contains the lateral intraparietal area. This latter area is involved in selectively attending to stimuli based on spatial location, and the intraparietal sulcus itself is said to contain a spatial attention ‘map’. Another such map of spatial attention is said to exist in the frontal eye fields [64].

The frontal eye fields (anterior cortex) and superior colliculus (midbrain) also play a role in selective attention, but this is often considered in terms of the voluntary control of eye movements [64]. This particular function is of less interest in the case of 3D-MOT as eyes are fixed on a given neutral point unrelated to the attentional component of the task. Along with the other anterior regions involved in the top-down aspects of selective attention, the frontal eye fields will be discussed in the section on inhibition and executive functions, both located below.

Beyond selective attention, the 3D-MOT task also places a significant demand on sustained attention. Sustained attention, popularly, but not always scientifically analogous to *vigilance*, is the capacity to maintain attention on a given stimulus for a given period of time, typically considered to be in the realm of 20 to 30 minutes [63, 65]. Neuroanatomically, sustained visual attention is associated with a wide range of brain regions, including the frontal, parietal and occipital cortices, the thalamus, the caudate and the terebellum [66].

The 3D-MOT task engages sustained attention in two ways. First, attention must be sustained on targets throughout the 8 seconds of the movement phases. Although brief in duration, this is a relatively taxing process as multiple dynamic objects are the targets of this attentive process. In our flashlight analogy, this is akin to splitting the beam onto multiple, distinct areas, and following a target with each. Second, because one series of the task consists of 20 trials and is repeated three times within each session, these attention abilities are required for a period of approximately 30 minutes [24].

This allocation of multifocal attention can be defined as distributed attention. Distributed attention is a dynamic process of allocating more or less attention to a given target based on its context; a target surrounded by ‘noise’ (to use 3D-MOT as an example, think of a target surrounded by multiple distractors [67]) requires more attention than a target which is relatively isolated. Because context is fluid and changing, this capacity must also be dynamic to be efficient and successful [68].

It may be noteworthy to distinguish between *distributed* and *divided* attention. Both refer to fractioning attentional resources to separate loci, necessarily plural, and thus both can be considered components of *multifocal* attention. Both distributed and divided attention can

be multimodal, meaning that attention can be allocated differentially across modalities, for example in visual and auditory domains. While the distinction between these two elements is still unclear in a general sense, we can better understand these facets of divided attention in the oft-cited context of *multitasking* [69].

To *multitask* essentially means doing two or more (*multi*) different things (*task*) at once. Hence, divided attention in this context can be operationally defined as the allocation of attentional resources to multiple loci in order to participate in *multiple* goal-oriented behaviours. Simply put, an individual is doing two things at once, such as driving and texting, and splitting their attention between them. Most consider multitasking to be brought about by technological evolution [70]. Some argue that it is a bit of a neuroscientific 'fad', and that multitasking is inherently negative [71, 72]. Others almost regard multitasking to be the ultimate panacea of human evolution and the quintessential goal of cognitive enhancement [73]. Others say it doesn't exist [74]. It is an impossible debate to settle, however there is ample neuroscientific evidence that multitasking is not always a desirable capacity [75].

Distributed attention, in contrast, can be defined in this context as the allocation of attentional resources to multiple loci in order to participate in a *singular* goal-oriented behaviour. Attention is still divided, but it is divided for the purposes of one specific task. This allocation of attention is inherent in a large number of contexts, but to remain in the realm of multitasking, the analogy of driving will be used.

While driving, attention must be divided between multiple sources of information; the driver, the vehicle, other vehicles, and the context (for example, back-country dirt road versus 10-lane superhighway). Attention will dynamically shift between the most important, or salient, elements. There is only one goal: driving. This is an example of the distribution of attention [68].

While driving, if sending a text message, changing radio stations, eating, or engaging in some other activity, there is suddenly a second goal. This demands a further fractioning of attention to a second goal-oriented task. This is the division of attention in the context of multitasking. Having a singular goal: the identification of target spheres, the standard 3D-MOT paradigm is thus a distributed attention task and not multitasking.

It is also possible to conceptualize visual attention as three interactive networks: *alerting*, achieving and maintaining an alert state; *orienting*, selecting information for sensory input; and *executive control*, resolving conflicting information [76]. These three facets of attention also lend themselves well to the 3D-MOT task. Alerting becomes the preparation of attentional resources prior to the task, during *preparation*, and continues during the identification phase via the heterogeneous salience of targets versus distractors. Orienting, then, concerns the maintenance of

attention on targets during task *performance*. Finally, the executive control portion contributes to the inhibition of distractors, and the modulation of attentional load applied dynamically to each target in the dynamic context.

As a final note on attention in the context of 3D-MOT, as Meyerhoff and colleagues [77] report: "the attentional processes studied with the multiple object paradigm apparently match the attentional processing during real-world tasks such as driving or team sports" making it an excellent candidate in the study of dynamic visual attention.

Inhibition

Inhibition is intrinsically linked with attention. To borrow the analogy first presented above, if attention is what is seen in the light of a flashlight, inhibition is the lens adjustment that illuminates a larger or smaller area, in lesser or greater detail. In this sense, attention is the 'positive' application of cognitive resources to a stimulus, while inhibition is the 'negative', a block on resources being diverted to a given stimulus [63].

In the 3D-MOT task, inhibition is applied to non-target spheres in order not to confound them with targets. As with attention, inhibition must be distributed dynamically across stimuli based on context; irrelevant information that is not proximal to a target is theoretically less important to inhibit than when proximal. As was mentioned above, an interesting feature of inhibition is that it appears less subject to the same limitations as attention, at least within the MOT task [36, 37, 40].

Inhibition plays an essential role in the attenuation of distractors and on-line updating of competing information between distractors and targets. Working in harmony with attention, these two processes are often considered an executive function [76].

Multiple analogies haven't been used by theoreticians to describe the relationship between attention and inhibition. Whether it is a screen and filter, flashlight and lens, or any other representation, the idea remains the same: attention is what 'gets through' – what is seen on the screen, in the light, or by the *conscious* – and inhibition is what does not – what gets blocked, remains in the dark, inaccessible to the conscious, and perhaps feeding the 'unconscious' [78].

Working memory

The ability to retain information for a short period of time is typically considered short-term memory. The manipulation and transformation of information held in an 'on-line' memory buffer is, in contrast, working memory [63]. A high-order function, working memory is often considered an executive function, and it draws upon short-term memory [79].

Both capacities are required in a 3D-MOT task [80]. Short-term memory is the less taxed of the two, in that it is necessary for the retention of targets versus distractors during the identification phase, and very early in the

movement phase before working memory comes online. In an extreme case of MOT in which movement is non-existent or imperceptible, the task would tax only short-term memory, and not significantly working memory.

As the movement phase initiates, the *tracking* – the transformation of this information, the identification of targets versus distractors moving through space-time – must be continuously updated and maintained throughout changes in direction, occlusion and other sources of competing information requiring an updated representation of targets versus distractors [81]. This information must also be held during the response phase, which is likely a shared task between working and short-term memory processes [79].

Working memory interacts with attention and inhibition to a large extent during the MOT task, and some research even shows that when working memory is taxed with multitasking, MOT performance suffers [82]. Visual attention and visual working memory could even share a neural substrate across anterior and posterior cortical regions, including the intraparietal sulcus [83].

Also, of note is the current trend of working memory training reviewed above (for example, see Westerberg and colleagues [84]). In spite of limited evidence of transfer [85], this type of training is often touted for its benefits with regard to executive functions, attention, and working memory itself [86], not to mention the claims of improvement in fluid intelligence [87] although some dismiss this [88].

Still, the interest in working memory in terms of cognitive enhancement is undeniable and understandable, and working memory enhancement does appear to increase activity in the prefrontal and parietal cortices [89], white matter volume [90], and the density of dopaminergic (D1) receptors [91]. For these reasons, the working memory component of the 3D-MOT task is potentially of great value from a cognitive-enhancement standpoint.

Executive functions

At the 'top end' of cognitive functions are found the executive functions. Planning, organizing, anticipating, decision making and self-regulating are all said to be among the classic executive functions [63].

Finally, in the 3D-MOT task executive functions are solicited by prediction of target movement, occlusion, as well as the modulation of attention and inhibition processes required to track targets amongst distractors [26, 44, 92]. Attention in the MOT task must be distributed dynamically; targets which are isolated require less 'attentiveness' than targets which are surrounded by distractors [54]. This dynamic allocation of attention is said to be mediated by executive functions [76, 93, 94, 95], and executive functions are also said to participate and help orchestrate the work of the 'where/how' visual stream described above [96].

Ecological validity

The theoretical argument for utilizing 3D-MOT as a cognitive enhancer can be addressed in four (4) main points that make it appropriate for use in this domain. These characteristics, centered on learning [23, 97] and ecological validity [77, 98], make it a candidate to improve performance with transfer.

- 1) The use of *multiple objects*, four targets and four distractors, mimics the multi-focal nature of everyday attention [99]. The dynamic nature of attention within and outside of the task also contributes to ecological validity [100].
- 2) The *large visual field* in which the task takes place (a 1.5 metre cube, see Method section for more details) draws on a wide attentional field [101]. Relevant and irrelevant attentional stimuli (targets and distractors) go well beyond the small, accurate space of the fovea, which also reflects real-world conditions [99].
- 3) The use of a *stereoscopic 3-dimensional environment* also reflects real-world conditions and allows for both collisions and occlusions [44]. Visual object tracking is better in a 3D rather than 2D environment [102].
- 4) The *adaptive staircase* regulating task difficulty is effective at keeping a trainee within their zone of proximale development, thus maximising learning and performance [103].

To summarize the review of 3D-MOT above: 3D-MOT can be considered a perceptual-cognitive training program. A trial in this task is brief, with the movement (training) phase lasting only 8 seconds, at the end of which a participant is provided with feedback. Repeatedly presented with this paradigm, participants demonstrate improvements in task performance, reflecting learning. As a cognitive enhancement paradigm, 3D-MOT has demonstrated efficacy in improving attention, working memory and information processing speed in healthy adults. Following training, functional brain changes as measured by qEEG have also been shown [24].

qEEG in MOT

A portion of the EEG neuroimaging literature related to MOT uses event-related potentials (ERPs), and this use of EEG is not readily relevant to review in the context of cognitive enhancement. An example of this work can be found in Doran and Hoffman [104] who examine the role of attention in the MOT task, or Drew and colleagues [105] who investigate the impact of the various MOT variables (speed and number of distractors) on task difficulty.

In terms of quantitative EEG (qEEG) literature, which is more easily adapted to the context of cognitive enhancement, little insight is available. One study reports an increase of beta power from the baseline to task condition seen generally across the cortex. Authors suggest that the right frontal cortex may be the most appropriate site for measuring cognitive load [106]. This last study is difficult to interpret namely because an eyes-closed

baseline is used as comparator against the task condition. Prior qEEG research has shown a difference in eyes-opened versus eyes-closed baselines [107], meaning that the differences observed in this study could simply be the result of a contrast between the eyes opened and eyes closed state. A doctoral dissertation on the topic of MOT shared similar results; an increase in beta/theta ratio during training over the bilateral frontal and occipital cortices [108].

Alpha activity is found to appear in tonic bursts over the bilateral inferior parietal cortex during a multiple object processing task, appearing more frequently when at “capacity limits” (defined by number of targets, in this case approximately 4) [109].

VanRullun [110] also notes a relationship between theta oscillations and attentional fluctuations within an object tracking and identification paradigm. VanRullun [110] reports that although Holcombe & Chen's [111] multiple object tracking findings do not include qEEG, they are potentially relatable. In a multiple object tracking paradigm, these authors discovered a slowing of attention sampling from 7, to 4 and finally to 2.6 Hz when tracking one, two and three objects respectively [111]. Attentional sampling of this type has been correlated with theta activity elsewhere [112]. Interestingly, a master's thesis used a MOT task as an outcome measure to demonstrate improved visual attention following 10 sessions of neurofeedback theta coherence training [113], further suggesting a strong link between theta and MOT.

Finally, a study by Bland and colleagues [114] highlights the role of gamma coherence in mediating interhemispheric integration and communication during MOT.

More studies combining 3D-MOT outcome measures with neuroimaging, including qEEG, would be beneficial and allow a better understanding of the neural mechanisms at play.

Additional considerations

In examining MOT literature, while there does not appear to be a ceiling effect for speed [39, 44], there is often a plateau that is observed in task performance; at this point improvements become quantitatively smaller over time. Evidence from clinical applications tends to demonstrate that when this ‘soft ceiling’ is reached, modifying the state of the trainee (for example, moving from a seated to standing position) can restart a learning and enhancement process. In the new state, performance initially suffers but increases once again with continued training.

Different populations also demonstrate different learning abilities in the 3D-MOT task. Professional athletes are better than amateurs, whom are in turn better than college students [51]. Continued training of amateur athletes could

potentially lead some to reach a professional level. The same could be possible for aging populations, who perform at a lower level than younger adults, but appear to learn at a similar rate [115]. More, it is possible that those with neurodevelopmental conditions [25], who perform at a lower level than their normal functioning peers, can be trained over time to perform at a similar level to those with no deficits. What consequence this will have on outcome measures for these populations is of primary interest and could demonstrate the clinical utility of this cognitive training tool.

Conclusion

A broad goal of cognitive enhancement is to enrich one's personal, academic, and professional life. Currently explored methods to bolster cognitive, emotional, and behavioural functions are as numerous as they are plagued by limitations including cost, side effects, ethical concerns, lack of transfer, and outright ineffectiveness [9].

As the 3D-MOT task is said to be ecologically valid, highly relevant to real-world capacities [77], implicated in visual information processing [53, 54], attention [31, 53, 67], inhibition [36, 37, 40], short-term and working memory [80], as well as executive functions [26, 44, 54, 92], the relevance of these results to other cognitive domains and applications is significant. Prior work has shown improvements in these domains in healthy adults, those with neurodevelopmental deficits [25], athletes [26], and aging populations [27, 28]. More studies for all of these populations, as well as others, are needed.

The goal of enhancing one's cognition, emotion and behaviour is inherent to humanity. As a population, we spend billions of dollars every year in this pursuit. Here, the arguments and available evidence supporting the use of 3D-MOT in this context is presented. While some promising peer-reviewed research exists, more investigations are needed to robustly establish the beneficial effects of this method in the context of cognitive enhancement. It is hoped that the framework provided herein will contribute to a better understanding of, and investigation into, the effects of 3D-MOT training on cognitive performance. ■

Conflicts of interest:

BP is a Scientific Advisor of CogniSens Athletics Inc. who produce the commercial version of the 3D-MOT program (NeuroTracker) presented in this manuscript. In this capacity, he holds shares in the company.

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